

AI Adoption Analytics

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Introduction



AI coding tools like Cursor and GitHub Copilot are widely adopted, yet organizations struggle to measure real usage and proficiency objectively.

Telemetry exists across multiple APIs, but data is fragmented, difficult to interpret, and rarely linked to competency frameworks or validated evidence.

Self-assessment and manual reporting often fail to capture authentic adoption patterns or detect superficial end-of-period usage.

AI Adoption Analytics transforms raw AI-tool telemetry into structured proficiency intelligence for engineering teams.

The platform integrates Cursor sync, Copilot metrics, evidence summarization, LLM profiles, vector-based scoring, and usage-pattern auditing.

It helps managers understand developer behaviour, model preferences, and competency alignment without deep technical expertise.

The system bridges the gap between tool dashboards and workforce planning at R Systems International.

Encourages data-driven AI adoption measurement and evidence-based talent development decisions.

Supports batch processing for entire engineering rosters with MySQL snapshot storage.

Methodology



The project begins with understanding enterprise needs for objective AI adoption measurement. Objectives are defined to focus on accuracy, automation, scalability, and usability. Cursor and Copilot API structures are studied to design reliable MySQL schemas. Data is collected from Cursor Admin API, Analytics API, and GitHub Copilot org metrics. Evidence files (PDF, DOCX, images) are collected per user for qualitative context. Data quality checks validate email linkage, date coverage, and row completeness. Telemetry, evidence, and Copilot data are merged into unified context JSON per user. LLM-based profiles are generated and embedded for competency similarity matching. A 7-day auditor agent analyses daily usage for concentration, spikes, and risk flags. Batch pipeline runs parallel workers for hundreds of team members. Streamlit UI provides interactive pipeline execution and result visualization. Outputs are reviewed via SQL snapshots and JSON artifacts for correctness. Iterative Agile cycles refine thresholds, error handling, and worker performance. The platform is tested under single-user, batch, and audit-only scenarios.

Tools and Technologies



Python : Main language for sync modules, agents, profile pipeline, and Streamlit UI.

MySQL : Stores Cursor daily usage, Copilot metrics, team roster, and proficiency snapshots.

Streamlit : Interactive web dashboard for pipeline runs, batch jobs, and audit viewing.

Cursor Admin & Analytics APIs : Fetch team members, daily usage, events, and insights.

GitHub Copilot Metrics API : Sync org-level seats and per-user daily usage reports.

Hugging Face LLM (Qwen-VL) : Multimodal profile generation and evidence summarization.

Pinecone + BGE Embeddings : Vector similarity matching against 43 competency items.

mysql-connector-python & requests : Database connectivity and HTTP API clients.

ThreadPoolExecutor : Parallel batch processing for multi-user pipeline execution.

auditor_agent.py : Rule-based 7-day usage pattern analysis with risk scoring.

Git / GitHub : Version control and collaborative development.

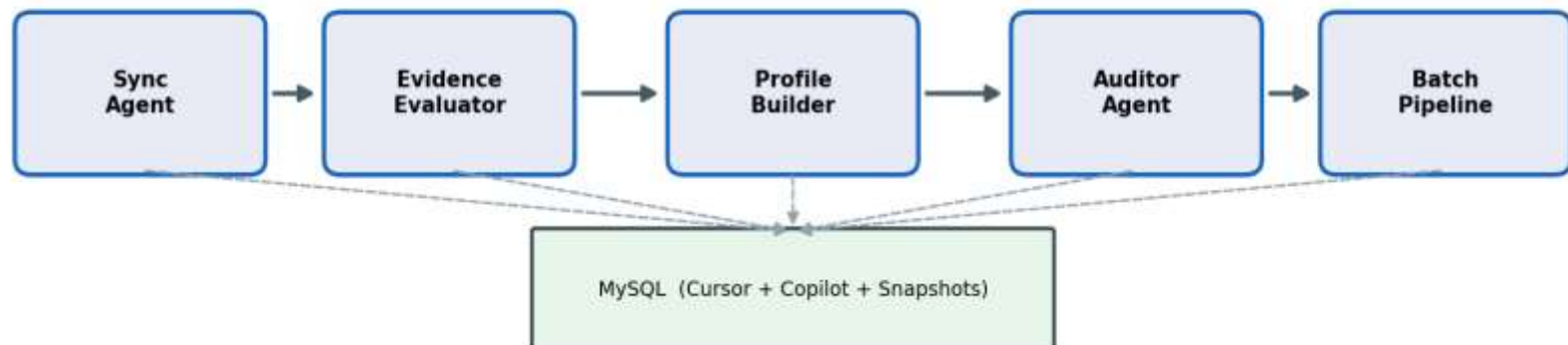
JSON Artifacts : Context, profile, scores, and audit outputs per user per period.

Flowchart



High-Level Pipeline Flow

Multi-Agent Architecture



Conclusion and Future Scope



Conclusion

AI Adoption Analytics successfully addresses the problem of fragmented AI-tool data and subjective adoption reporting.

The project unifies Cursor telemetry, Copilot metrics, and evidence into actionable proficiency insights.

By combining data engineering, LLMs, vector search, and usage auditing, the platform supports informed workforce decisions.

Overall, the project bridges the gap between raw tool metrics and structured competency assessment.

Future Scope

Extend auditor agent to include Copilot daily usage and persist audit snapshots in MySQL.

Executive dashboards for org-wide adoption trends and level distribution.

Real-time scheduled sync and automatic daily snapshot refresh.

RAG over evidence for deeper competency-to-artifact linking.

Human reviewer workflow for official framework certification.

Predictive models for forecasting adoption trajectories and training needs.

References



Streamlit Documentation: Streamlit is an open-source Python framework used to build interactive data dashboards and web applications quickly. It is widely used for data analysis and visualization projects. The official Streamlit documentation provides clear guides, examples, and API references for developers. **Visit Streamlit Documentation :** <https://docs.streamlit.io/>

Pandas Documentation: Pandas is a powerful Python library used for data manipulation and analysis. It provides data structures like DataFrames to clean, transform, and analyze structured data efficiently. The Pandas documentation explains data handling techniques, preprocessing methods, and real-world examples. **Visit Pandas Documentation :**

<https://pandas.pydata.org/docs/>

NumPy Documentation:

NumPy is a fundamental Python library for numerical computing. It supports large multi-dimensional arrays and mathematical operations, making it essential for financial data processing. The NumPy documentation offers detailed explanations of array operations and numerical functions. **Visit NumPy Documentation :** <https://numpy.org/doc/>

Plotly Documentation: Plotly is an interactive data visualization library used to create dynamic charts and graphs. It helps present financial insights in an intuitive and user-friendly manner. The Plotly documentation provides tutorials, examples, and customization options. **Visit Plotly Documentation :** <https://plotly.com/python/>



*Thank
you!*

